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|----------------------------------------------------|----------------------------------------------------------------------------|
| Program: <b>Diploma in Artificial Intelligence</b> |                                                                            |
| Course Code: <b>5381</b>                           | Course Title: <b>Fundamentals of Virtual Reality and Augmented Reality</b> |
| Semester: <b>5</b>                                 | Credits: <b>4</b>                                                          |
| Course Category: <b>Program Core</b>               |                                                                            |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>           | Periods per semester: <b>60</b>                                            |

### Course Objectives:

- Outline principles of geometric modeling, transformations, and generic VR systems.
- Demonstrate proficiency in animating objects, creating transitions, and simulating physical interactions in virtual environments.
- Understand the components, functionalities, and applications of augmented reality (AR) systems within VR contexts.

### Course Prerequisites:

| Course code | Course name                             | Semester |
|-------------|-----------------------------------------|----------|
|             | Introduction to Artificial Intelligence | 3        |

### Course Outcomes

On completion of the course, the students will be able to:

| CO <sub>n</sub> | Description                                                                                                       | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Understand Virtual Reality Fundamentals                                                                           | 14               | Understanding   |
| CO2             | Outline geometric modeling, transformations, and generic VR system principles.                                    | 15               | Understanding   |
| CO3             | Demonstrate animation of objects, create transitions, and simulate physical interactions in virtual environments. | 15               | Understanding   |
| CO4             | Understand AR system components, functionalities and applications.                                                | 14               | Understanding   |
|                 | Series Test                                                                                                       | 2                |                 |

## CO-PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Description                                                                           | Duration (Hours) | Cognitive Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Understand Virtual Reality Fundamentals</b>                                        |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Describe the concept of Virtual Reality (VR) and Virtual Environment                  | 3                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Outline the historical development of Virtual Reality                                 | 2                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Explain the principles of 3D computer graphics                                        | 4                | Understanding   |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Illustrate basic 3D modeling techniques to create virtual objects and environments.   | 5                | Understanding   |
| <b>Contents:-</b> Virtual Reality & Virtual Environment: Introduction – Computer graphics – Real time computer graphics –Flight Simulation – Virtual environments –requirement – benefits of virtual reality<br><br>Historical development of VR : Scientific Landmark.<br><br>3D Computer Graphics: The Virtual world space – positioning the virtual observer – the perspective projection – human vision – stereo perspective projection – 3D clipping – Colour theory – Simple 3D modeling– Illumination models – Reflection models – Shading algorithms- Radiosity – Hidden Surface Removal – 1-Stereographic image. |                                                                                       |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Outline geometric modeling, transformations, and generic VR system principles.</b> |                  |                 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                            |   |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---|---------------|
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                                            | Describe the transition from 2D to 3D geometric modeling techniques used in Virtual Environments (VE).                     | 3 | Understanding |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                            | Illustrate frames of reference and modeling transformations to create and manipulate objects within VR.                    | 4 | Understanding |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                            | Identify techniques such as instances, picking, flying, scaling the VE, and collision detection.                           | 4 | Understanding |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                                            | Outline the components of a generic VR system, including the virtual environment, computer environment, and VR technology. | 4 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Series Test – I                                                                                                            | 1 |               |
| <b>Contents:</b><br>Geometric Modeling: From 2D to 3D – 3D space curves – 3D boundary representation<br>Geometrical Transformations: Frames of reference – Modeling transformations – Instances – Picking – Flying – Scaling the VE – Collision detection<br>Generic VR system: The virtual environment – the Computer environment – VR Technology – Model of interaction – VR Systems                                                           |                                                                                                                            |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Demonstrate animation of objects, create transitions, and simulate physical interactions in virtual environments.</b>   |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                                            | Illustrate linear and non-linear interpolation methods to animate objects.                                                 | 3 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                                            | Explain animation techniques for translating objects both linearly and nonlinearly.                                        | 3 | Understanding |
| M3.03                                                                                                                                                                                                                                                                                                                                                                                                                                            | Demonstrate shape and object inbetweening techniques to create transitions                                                 | 3 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                            | Describe the methods of Physical Simulation in Virtual Environments                                                        | 3 | Understanding |
| M3.05                                                                                                                                                                                                                                                                                                                                                                                                                                            | Outline elastic collision simulations to objects                                                                           | 3 | Understanding |
| <b>Contents:</b><br>Animating the Virtual Environment: The dynamics of numbers – Linear and Non-linear interpolation - The animation of objects – linear and nonlinear translation - shape & object inbetweening – free-from deformation – particle system.<br>Physical Simulation : Objects falling in a gravitational field – Rotating wheels – Elastic collisions – projectiles – simple pendulum – springs – Flight dynamics of an aircraft. |                                                                                                                            |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Understand AR system components, functionalities and applications.</b>                                                  |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                            | Explain AR Taxonomy and Technologies                                                                                       | 2 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                            | Differentiate AR and VR                                                                                                    | 2 | Understanding |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                |   |               |
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| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outline the components and functionalities of augmented reality (AR) systems.                                  | 2 | Understanding |
| M4.04                                                                                                                                                                                                                                                                                                                                                                                                                                              | Outline tracking methods, visualization techniques, and interaction models.                                    | 3 | Understanding |
| M4.05                                                                                                                                                                                                                                                                                                                                                                                                                                              | Explain advanced augmented reality (AR) methods such as marker-less tracking and mobile projection interfaces. | 3 | Understanding |
| M4.06                                                                                                                                                                                                                                                                                                                                                                                                                                              | Summarize strategies to enhance interactivity and user engagement within augmented reality environments.       | 2 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Series Test – II                                                                                               | 1 |               |
| <b>Contents:</b><br>Taxonomy, technology and features of augmented reality, difference between AR and VR, Challenges with AR, AR systems and functionality, Augmented reality methods, visualization techniques for augmented reality, wireless displays in educational augmented reality applications Mobile projection interfaces, marker-less tracking for augmented reality, enhancing interactivity in AR environments, evaluating AR systems |                                                                                                                |   |               |

**Text / Reference:**

| T/R | Book Title / Author                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------|
| T1  | Virtual Reality Systems, John Vince, Pearson Education Asia, 2007                                                                              |
| T2  | Augmented and Virtual Reality, Anand R, Khanna Publishing House, Delhi.                                                                        |
| R1  | Developing Virtual Reality Applications, Alan Craig, William Sherman and Jeffrey Will, Foundations of Effective Design, Morgan Kaufmann, 2009. |
| R2  | Virtual Reality Technology, Grigore C. Burdea, Philippe Coiffet, Wiley 2016                                                                    |

**Online Resources:**

| Sl. No. | Website Link                                                                                                                                                                                                |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <a href="https://www.simplilearn.com/tutorials/artificial-intelligence-tutorial/what-is-virtual-reality">https://www.simplilearn.com/tutorials/artificial-intelligence-tutorial/what-is-virtual-reality</a> |
| 2       | <a href="https://www.youtube.com/watch?v=YBQ_ps6e71k">https://www.youtube.com/watch?v=YBQ_ps6e71k</a>                                                                                                       |
| 3       | <a href="https://www.softwaretestinghelp.com/what-is-augmented-reality/">https://www.softwaretestinghelp.com/what-is-augmented-reality/</a>                                                                 |